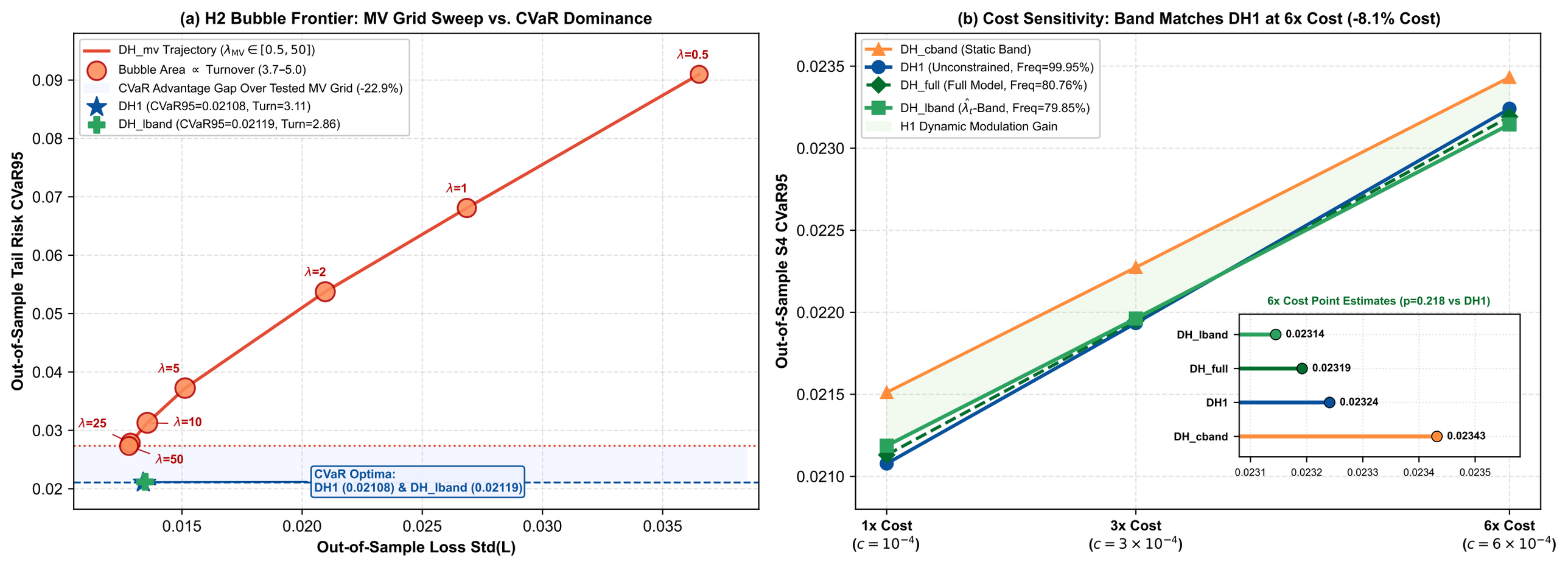

(a) H2 Bubble Frontier: MV Grid Sweep vs. CVaR Dominance
DH_mv Trajectory ($\lambda_{MV} \in [0.5, 50]$)
Bubble Area ∝ Turnover (3.7–5.0)
CVaR Advantage Gap Over Tested MV Grid (-22.9%)
DH1 (CVaR95=0.02108, Turn=3.11)
DH_lband (CVaR95=0.02119, Turn=2.86)
λ=0.5
λ=1
λ=2
λ=5
λ=10
λ=25
λ=50
CVaR Optima:
DH1 (0.02108) & DH_lband (0.02119)
Out-of-Sample Tail Risk CVaR95
Out-of-Sample Loss Std(L)
0.02
0.03
0.04
0.05
0.06
0.07
0.08
0.09
0.015
0.020
0.025
0.030
0.035
(b) Cost Sensitivity: Band Matches DH1 at 6x Cost (-8.1% Cost)
DH_cband (Static Band)
DH1 (Unconstrained, Freq=99.95%)
DH_full (Full Model, Freq=80.76%)
DH_lband ($\hat{\lambda}_t$-Band, Freq=79.85%)
H1 Dynamic Modulation Gain
Out-of-Sample S4 CVaR95
0.0210
0.0215
0.0220
0.0225
0.0230
0.0235
1x Cost
($c = 10^{-4}$)
3x Cost
($c = 3 \times 10^{-4}$)
6x Cost
($c = 6 \times 10^{-4}$)
6x Cost Point Estimates (p=0.218 vs DH1)
DH_lband
0.02314
DH_full
0.02319
DH1
0.02324
DH_cband
0.02343
0.0231
0.0232
0.0233
0.0234
0.0235